APPLICATIONS OF INTEGRATION

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Department of Mathematics
Kwame Nkrumah University of Science and Technology



Lecture Outline

1. Area of a Region Between Two Curves
2. Volume of A solid
   - Disk Method
   - Washer Method
   - Volume of A solid With Known Cross Sections
3. Arc Length and Surfaces of Revolution
4. Work and Fluid Forces



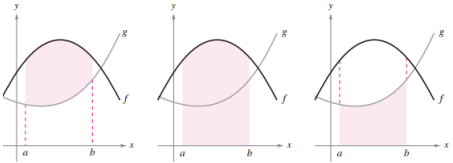
Outline of Presentation

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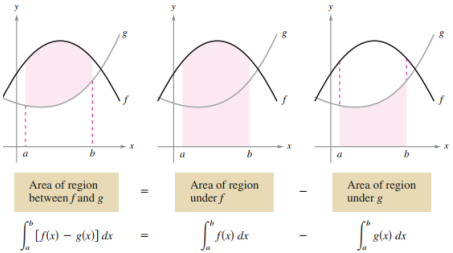
Area of a Region Between Two Curves

We you can extend the application of definite integrals from the area of a region under a curve to the area of a region between two curves. Consider two functions and that are continuous on the interval $[a, b]$



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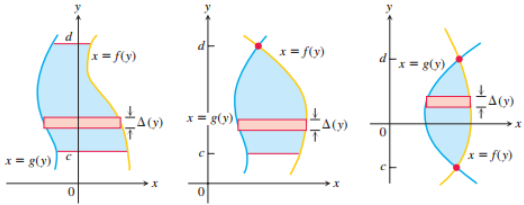


If f and g are continuous on $[a,\ b]$ and $g(x) \le f(x)$ for all x in $[a,\ b]$ then the area of the region bounded by the graphs of f and g and the vertical lines $x = a$ and $x = b$ is

$$A = \int_a^b [f(x) - g(x)]\,dx = \int_{a=x_1}^{b=x_2} [\text{top curve - bottom curve}]\,dx$$

Area of a Region Between Two Curves

If a region's bounding curves are described by functions of y, the approximating rectangles are horizontal instead of vertical and the basic formula has y in place of x.



$$A = \int_c^d [f(y) - g(y)]\,dy = \int_{c=y_1}^{d=y_2} [\text{right curve - left curve}]\,dy$$



Area of a Region Between Two Curves

Example

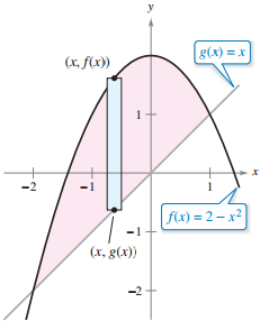
Find the area of the region bounded by the graphs of $f(x) = 2 - x^2$ and $g(x) = x$



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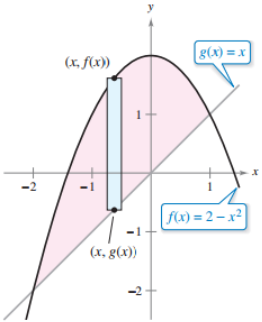
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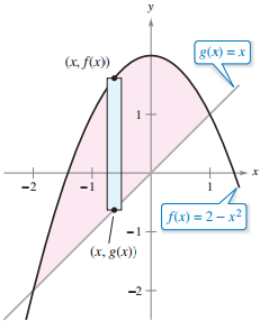
The plot have two intersections points; that is the limits for integration. First equate $f(x)$ to $g(x)$ to find these points.

$$2 - x^2 = x \implies x^2 + x - 2 = 0 \implies x = -2, \text{ or } 1 \tag{1}$$

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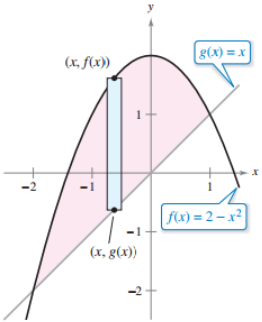
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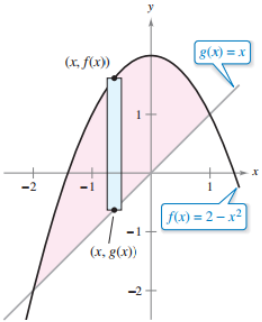
$$= \int_{-2}^{1} [(2 - x^2) - x]\,dx \tag{3}$$



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$$= -\frac{x^3}{3} - \frac{x^2}{2} + 2x \Big|_{-2}^{1} = \frac{9}{2} \tag{4}$$

Area of a Region Between Two Curves

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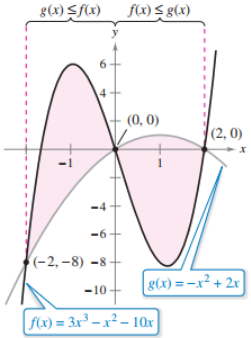
Find the area of the region between the graphs of $f(x) = 3x^3 - x^2 - 10x$ and $g(x) = -x^2 + 2x$



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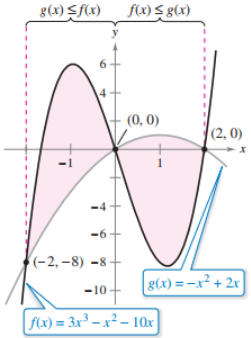
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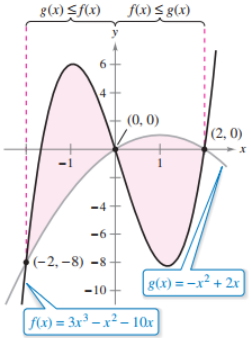
$$3x^3 - x^2 - 10x = -x^2 + 2x \implies 3x^3 - 12x = 0$$

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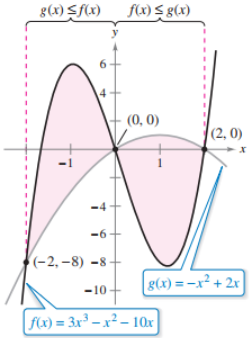
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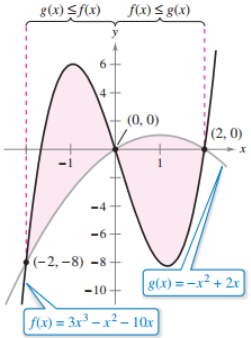
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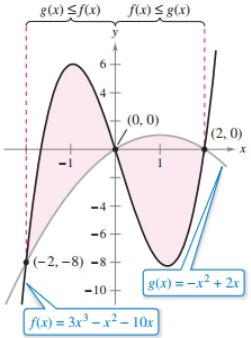
$$A = \int_{-2}^{0} [f(x) - g(x)]\,dx + \int_{0}^{2} [g(x) - f(x)]\,dx \tag{5}$$



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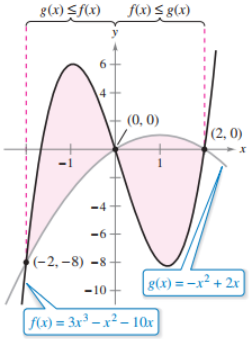
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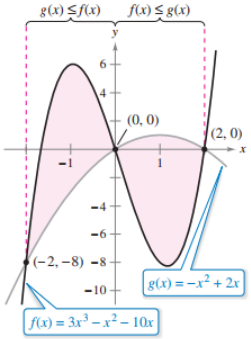
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Area of a Region Between Two Curves

Exercise

1. Find the areas of the regions enclosed by the following curves
   1. $y = x^2 - 2$ and $y = 2$
   2. $y = 2x - x^2$ and $y = -3$
   3. $y = x^4$ and $y = 8x$
   4. $y = 7 - 2x^2$ and $y = x^2 + 4$
   5. $x = 2y^2, x = 0$, and $y = 3$
   6. $y^2 - 4x = 4$ and $4x - y = 16$
   7. $x + y^2 = 0$ and $x + 3y^2 = 2$



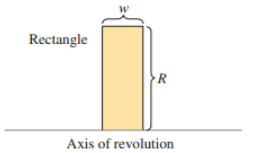
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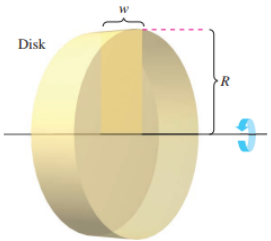
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Volume of A solid: Disk Method

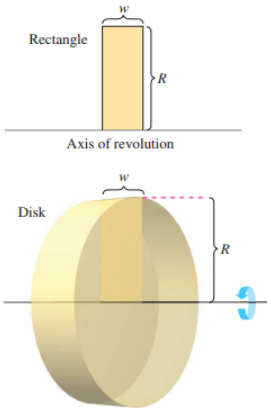
When a region in the plane is revolved about a line, the resulting solid is a solid of revolution, and the line is called the axis of revolution.





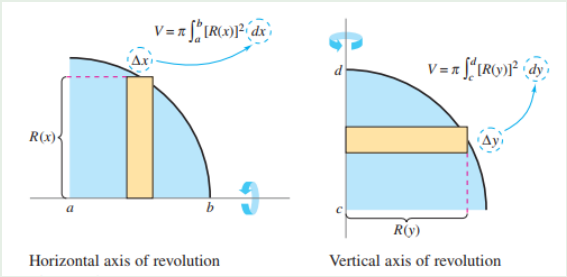
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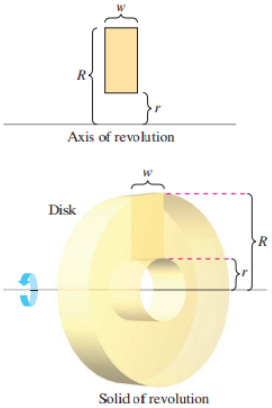
To find the volume of a solid of revolution with the disk method, use one of the formulas below:

$$V = \pi \int_a^b [R(x)]^2 dx \text{ or } V = \pi \int_c^d [R(y)]^2 dy$$



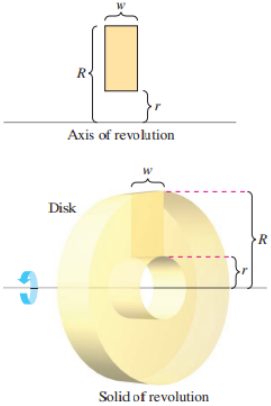
Volume of A solid: Washer Method

The disk method can be extended to cover solids of revolution with holes by replacing the representative disk with a representative washer. The washer is formed by revolving a rectangle about an axis with inner $r(x)$ and outer radii $R(x)$.



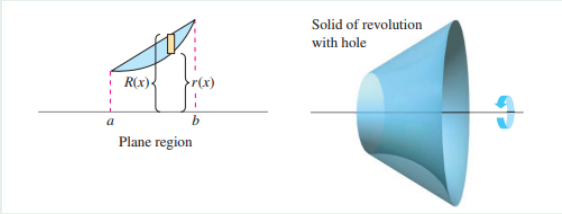
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The volume of a solid of revolution using the washer method is:

$$V = \pi \int_a^b \left[[R(x)]^2 - [r(x)]^2 \right] dx$$

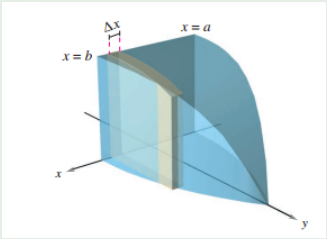


Volume of A solid With Known Cross Sections

With the disk method, you can find the volume of a solid having a circular cross section. This method can be generalized to solids of any shape, as long as you know a formula for the area of an arbitrary cross section.

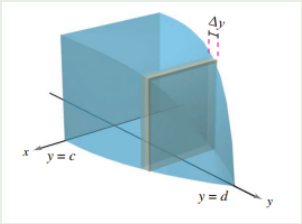
For cross sections of area $A(x)$ taken perpendicular to the x-axis

$$V = \int_a^b A(x)\,dx$$



For cross sections of area $A(y)$ taken perpendicular to the y-axis

$$V = \int_c^d A(y)\,dy$$



Volume of A solid

Example

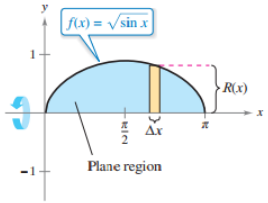
Find the volume of the solid formed by revolving the region bounded by the graph of $f(x) = \sqrt{\sin x}$ and the x-axis $x \in [0,\ \pi]$ about the x-axis.

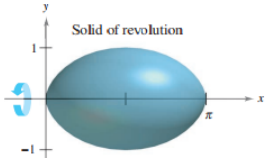


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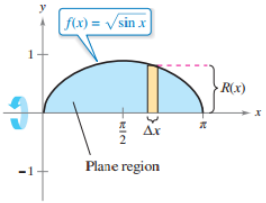


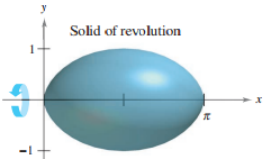


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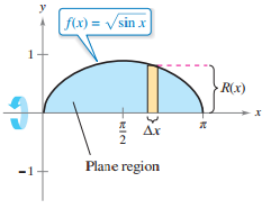


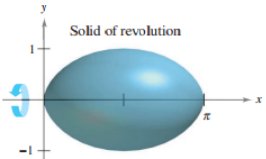
$$V = \pi \int_a^b [R(x)]^2 dx \tag{8}$$

Volume of A solid

Example

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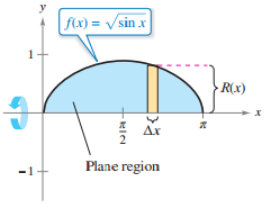
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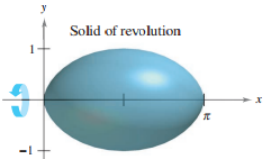
$$= \pi \int_0^\pi [\sqrt{\sin x}]^2 dx \tag{9}$$

Volume of A solid

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Find the volume of the solid formed by revolving the region bounded by the graph of $f(x) = \sqrt{\sin x}$ and the x-axis $x \in [0, \pi]$ about the x-axis.





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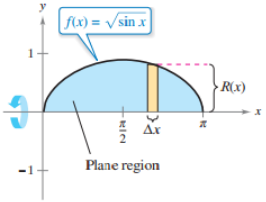
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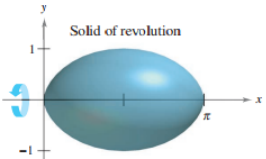
$$= \pi \int_0^\pi \sin x \, dx \tag{10}$$

Volume of A solid

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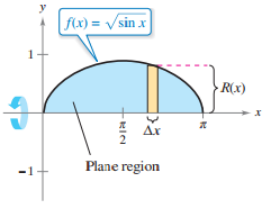
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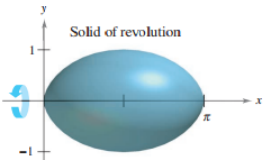
$$= \pi(-\cos x)\Big|_0^\pi \tag{11}$$

Volume of A solid

Example

Find the volume of the solid formed by revolving the region bounded by the graph of $f(x) = \sqrt{\sin x}$ and the x-axis $x \in [0,\ \pi]$ about the x-axis.





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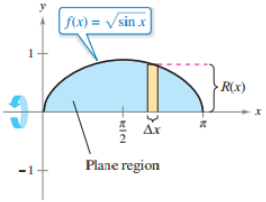
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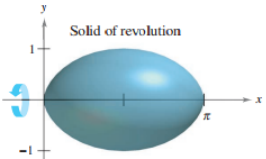
$$= \pi(1+1) \tag{12}$$

Volume of A solid

Example

Find the volume of the solid formed by revolving the region bounded by the graph of $f(x) = \sqrt{\sin x}$ and the x-axis $x \in [0,\ \pi]$ about the x-axis.





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$$= 2\pi \tag{13}$$

Volume of A solid

Example

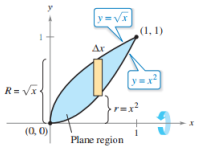
Find the volume of the solid formed by revolving the region bounded by the graphs of $y = \sqrt{x}$ and $y = x^2$ about the x-axis.

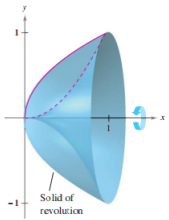


Volume of A solid

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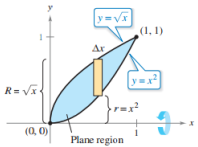


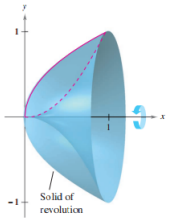


Volume of A solid

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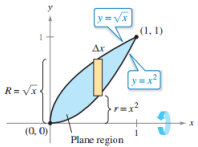


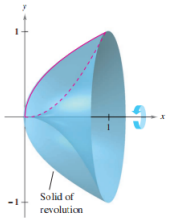
$$V = \pi \int_a^b \left[[R(x)]^2 - [r(x)]^2 \right] dx \tag{14}$$

Volume of A solid

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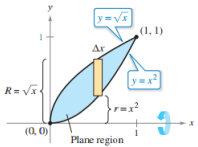
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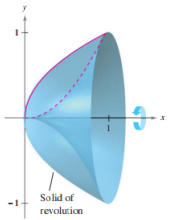
$$= \pi \int_0^1 \left[(\sqrt{x})^2 - (x^2)^2 \right] dx \tag{15}$$

Volume of A solid

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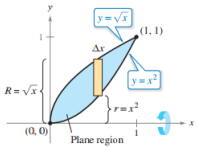
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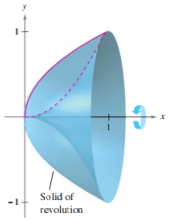
$$= \pi \int_0^1 (x - x^4) dx \tag{16}$$

Volume of A solid

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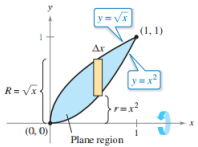
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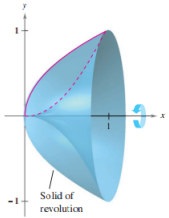
$$= \pi \left[\frac{x^2}{2} - \frac{x^5}{5} \right] \Big|_0^1 \tag{17}$$

Volume of A solid

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Find the volume of the solid formed by revolving the region bounded by the graphs of $y = \sqrt{x}$ and $y = x^2$ about the x-axis.





$$V = \pi \int_a^b \left[[R(x)]^2 - [r(x)]^2 \right] dx \tag{14}$$

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$$= \frac{3\pi}{10} \tag{18}$$

Volume of A solid

Example

Prove that the volume of a pyramid with a square base is $V = \frac{1}{3}hB$ where h is the height of the pyramid and B is the area of the base.

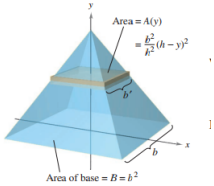


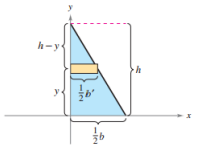
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As shown, you can intersect the pyramid with a plane parallel to the base at height y to form a square cross section whose sides are of length b'.



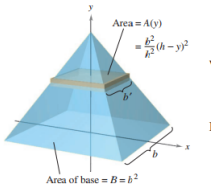


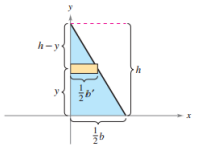
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Using similar triangles $\frac{b'}{b} = \frac{h-y}{h} \implies b' = \frac{b}{h}(h-y)$ so the Area of the square cross section is $A(y) = (b')^2 = \frac{b^2}{h^2}(h-y)^2$

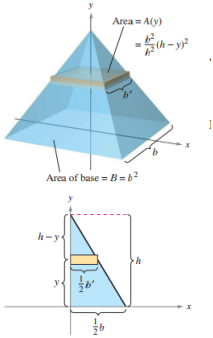


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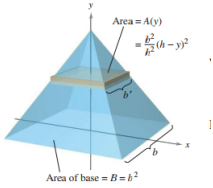
$$V = \int_0^h A(y)\,dy = \int_0^h \frac{b^2}{h^2}(h-y)^2\,dy \tag{19}$$

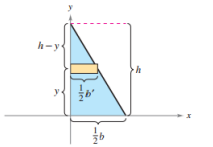
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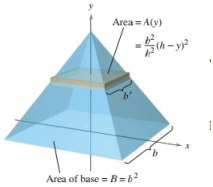
$$= \frac{b^2}{h^2}\int_0^h (h-y)^2\,dy = \frac{b^2}{h^2}\left[\frac{(h-y)^3}{3}\right]\Bigg|_0^h \tag{20}$$

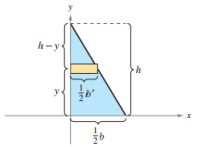
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$$V = \int_0^h A(y)\,dy = \int_0^h \frac{b^2}{h^2}(h-y)^2\,dy \tag{19}$$

$$= \frac{b^2}{h^2}\int_0^h (h-y)^2\,dy = \frac{b^2}{h^2}\left[\frac{(h-y)^3}{3}\right]\Bigg|_0^h \tag{20}$$

$$= \frac{b^2}{h^2}\left[\frac{h^3}{3}\right] = \frac{1}{3}hB, \quad \text{where } B = b^2 \tag{21}$$

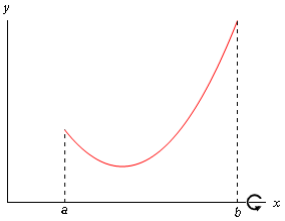
Exercise

1. Find the volume of the solid generated by revolving the region bounded by the graphs of the equations about the x-axis
   1. $y = \frac{1}{x},\ y = 0,\ x = 1,\ x = 3$
   2. $y = e^{-x},\ y = 0,\ x = 0,\ y = 1$
   3. $y = x^2 + 1,\ y = -x^2 + 2x + 5,\ x = 0,\ x = 3$
2. Find the volume of the solid generated by revolving the region bounded by the graphs of the equations about the y-axis
   1. $y = 3(2 - x),\ y = 0,\ x = 0$
   2. $y = 9 - x^2,\ y = 0,\ x = 2,\ x = 3$
3. Use the disk method to verify that the volume of a sphere is $\frac{4}{3}\pi r^3$, where r is the radius.



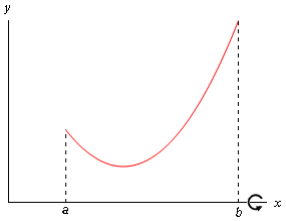
Arc Length and Surfaces of Revolution

To determine the length of the continuous function $y = f(x)$ on the interval $[a, b]$. We'll also need to assume that the derivative is continuous on $[a, b]$.



Arc Length and Surfaces of Revolution

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The length of this curve is given as

$$L = \int ds \tag{22}$$

where,

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \qquad \text{if } y = f(x),\ a \le x \le b \tag{23}$$

$$ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \qquad \text{if } x = h(y),\ c \le y \le d \tag{24}$$

Using the first ds will require x limits of integration and using the second ds will require y limits of integration. The denominator of the derivative is always the derivative of the integral.



Outline of Presentation

1. Area of a Region Between Two Curves
2. Volume of A solid
   - Disk Method
   - Washer Method
   - Volume of A solid With Known Cross Sections
3. Arc Length and Surfaces of Revolution
4. Work and Fluid Forces



Arc Length and Surfaces of Revolution

Example

Determine the length of $y = \ln(\sec x)$; where $0 \leq x \leq \frac{\pi}{4}$



Arc Length and Surfaces of Revolution

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$$y = f(x) = \ln(\sec x) \implies \frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x \implies \left(\frac{dy}{dx}\right)^2 = \tan^2 x$$



Arc Length and Surfaces of Revolution

Example

Determine the length of $y = \ln(\sec x)$; where $0 \le x \le \frac{\pi}{4}$

$$y = f(x) = \ln(\sec x) \implies \frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x \implies \left(\frac{dy}{dx}\right)^2 = \tan^2 x$$

Therefore

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \tan^2 x} = \sqrt{\sec^2 x} = \sec x$$



Arc Length and Surfaces of Revolution

Example

Determine the length of $y = \ln(\sec x)$; where $0 \le x \le \frac{\pi}{4}$

$$y = f(x) = \ln(\sec x) \implies \frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x \implies \left(\frac{dy}{dx}\right)^2 = \tan^2 x$$

Therefore

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \tan^2 x} = \sqrt{\sec^2 x} = \sec x$$

The arc length is then,

$$L = \int_0^{\frac{\pi}{4}} \sec x \, dx$$



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$$\begin{aligned} L &= \int_0^{\frac{\pi}{4}} \sec x \, dx \\ &= \ln|\sec x + \tan x| \Big|_0^{\frac{\pi}{4}} \\ &= \ln\left(\sqrt{2} + 1\right) \end{aligned}$$



Arc Length and Surfaces of Revolution

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Determine the length of $x = \frac{1}{2}y^2$; for $0 \leq x \leq \frac{1}{2}$. Assume that y is positive.



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Arc Length and Surfaces of Revolution

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$$\frac{dx}{dy} = y \qquad \Rightarrow \qquad \sqrt{1+\left(\frac{dx}{dy}\right)^2} = \sqrt{1+y^2}$$

Before writing down the length notice that we were given x limits and we will need y limits for this ds. With the assumption that y is positive these are easy enough to get. All we need to do is plug x into our equation and solve for y. Doing this gives,

$$0 \le y \le 1$$

The integral for the arc length is then,

$$L = \int_0^1 \sqrt{1+y^2}\, dy$$



This integral will require the following trigonometry substitution.

$$y = \tan\theta \qquad dy = \sec^2\theta\, d\theta$$

$$y = 0 \quad \Rightarrow \quad 0 = \tan\theta \quad \Rightarrow \quad \theta = 0$$

$$y = 1 \quad \Rightarrow \quad 1 = \tan\theta \quad \Rightarrow \quad \theta = \frac{\pi}{4}$$

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The length is then,

$$\begin{aligned} L &= \int_0^{\frac{\pi}{4}} \sec^3\theta\, d\theta \\ &= \frac{1}{2}\left(\sec\theta\tan\theta + \ln|\sec\theta+\tan\theta|\right)\Big|_0^{\frac{\pi}{4}} \end{aligned}$$



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Arc Length and Surfaces of Revolution

In this section we are going to look at solids of revolution. Below is a sketch of a function $y = f(x)$ in the interval $[a, b]$ and the solid of revolution we get by rotating the function about the x-axis.



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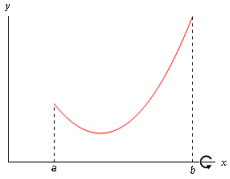


Figure A

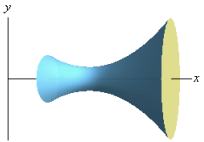


Figure B

Arc Length and Surfaces of Revolution

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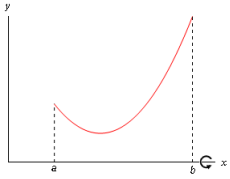


Figure A

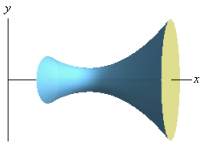


Figure B

The surface area of this solid (Figure B) is obtained using the formulas:

$$S = \int 2\pi y \, ds \qquad \text{rotation about } x-\text{axis} \tag{25}$$

$$S = \int 2\pi x \, ds \qquad \text{rotation about } y-\text{axis} \tag{26}$$

where,

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \qquad \text{if } y = f(x),\ a \le x \le b \tag{27}$$

$$ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \qquad \text{if } x = h(y),\ c \le y \le d \tag{28}$$

Arc Length and Surfaces of Revolution

Example

Determine the surface area of the solid obtained by rotating $y = \sqrt{9 - x^2}$, $-2 \leq x \leq 2$ about the x-axis.



Arc Length and Surfaces of Revolution

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The formula that we'll be using here is, $S = \int 2\pi y \, ds$

$$\frac{dy}{dx} = \frac{1}{2}\left(9 - x^2\right)^{-\frac{1}{2}}(-2x) = -\frac{x}{\left(9 - x^2\right)^{\frac{1}{2}}}$$



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Before evaluating the integral we'll need to substitute in for y as well.
The surface area is then,

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The surface area is then,

$$S = \int_{-2}^{2} 2\pi \sqrt{9 - x^2} \frac{3}{\sqrt{9 - x^2}} \, dx$$

$$= \int_{-2}^{2} 6\pi \, dx = 24\pi$$



Exercise

1. Determine the length of
$$y = \sqrt{x+2}, \qquad 1 \le x \le 7$$
using,
   1. $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
   2. $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$
2. Determine the length of $y = 7(6+x)^{\frac{3}{2}}, \qquad 189 \le y \le 875.$
3. Determine the length of $x = 4(3+y)^2, \qquad 1 \le y \le 4.$
4. Find the surface area of the object obtained by rotating
$$y = 4 + 3x^2, \qquad 1 \le x \le 2$$
about the y-axis.
5. Find the surface area of the object obtained by rotating
$$y = \sin(2x), \qquad 0 \le x \le \frac{\pi}{8}$$

Outline of Presentation

1. Area of a Region Between Two Curves
2. Volume of A solid
   - Disk Method
   - Washer Method
   - Volume of A solid With Known Cross Sections
3. Arc Length and Surfaces of Revolution
4. Work and Fluid Forces



Work and Fluid Forces

The work done by a variable force $F(x)$ in moving an object along the x-axis from $x = a$ to $x = b$ is

$$W = \int_a^b F(x)dx$$

Hooke's Law

Hooke's law says that the force required to hold a stretched or compressed spring x units from its natural (unstressed) length is proportional to x. In symbols $F = kx$



Work and Fluid Forces

Example

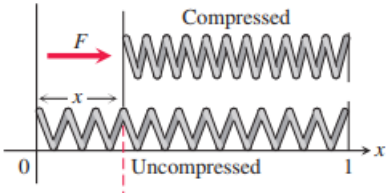
Find the work required to compress a spring from its natural length of $1ft$ to a length of $0.75ft$ if the force constant is $k = 16$.



Work and Fluid Forces

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Find the work required to compress a spring from its natural length of $1ft$ to a length of $0.75ft$ if the force constant is $k = 16$.



The work done by F over this interval is

$$W = \int_0^{0.25} 16x\,dx = 8x^2\Big|_0^{0.25} = 0.5$$

Exercise

1. A force of 5 pounds compresses a 15-inch spring a total of 3 inches. How much work is done in compressing the spring 7 inches?
2. A force of 250 newtons stretches a spring 30 centimeters. How much work is done in stretching the spring from 20 centimeters to 50 centimeters?
3. A force of 20 pounds stretches a spring 9 inches in an exercise machine. Find the work done in stretching the spring 1 foot from its natural position.
4. An overhead garage door has two springs, one on each side of the door. A force of 15 pounds is required to stretch each spring 1 foot. Because of the pulley system, the springs stretch only one-half the distance the door travels. The door moves a total of 8 feet, and the springs are at their natural length when the door is open. Find the work done by the pair of springs.



END OF LECTURE

THANK YOU

